Lecture 7 (Extreme Point, Vertex, and BFS Equivalence)

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We will define three related concepts for polyhedra, then show they are all equivalent. Translating between the language of polyhedral geometry, convexity, and LP optimization allows us to use whichever definition suits us for a given proof. First, some definitions:

Extreme Points

Definition

Given a polyhedron, $P = \left\{\vec{x} \in \mathbb{R}^n | A\vec{x} \geq \vec{b}\right\}$, we say $\vec{x} \in P$ is an **extreme point** of P if there does not exist $\lambda \in [0,1]$, $\vec{y}, \vec{z} \in P$ such that $\vec{y} \neq \vec{x}$, $\vec{z} \neq \vec{x}$, and

$$\vec{x} = \lambda\vec{y} + (1-\lambda)\,\vec{z}$$

Remarks

1. This is equivalent to saying an extreme point $\vec{x}$ cannot be written as a convex combination of points in $P - \{\vec{x}\}$.
2. Since we are assuming $\vec{y} \neq \vec{x}$, $\vec{z} \neq \vec{x}$, we often will assume $\lambda \in (0,1)$ when proving things about extreme points. This is because we already are assuming $\vec{y}$ and $\vec{z}$ are not equal to $\vec{x}$, and the convex combination with $\lambda \in \{0,1\}$ will yield $\vec{y}$ or $\vec{z}$
3. This definition can be extended to other convex sets and still makes sense. It does not make sense for sets that are not convex however.

Vertices

Definition

Given a polyhedron, $P = \left\{\vec{x} \in \mathbb{R}^n | A\vec{x} \geq \vec{b}\right\}$, we say $\vec{x} \in P$ is an **vertex** of P if $\exists \vec{c} \in \mathbb{R}^n$ such that

$$\vec{c} \cdot \vec{x} < \vec{c} \cdot \vec{y}$$

for all $\vec{y} \in P$ such that $\vec{y} \neq \vec{x}$.

Remarks

1. This implies that $\vec{x}$ is the unique optimal solution of some LP with feasible region, P.
2. This does not imply that $\vec{c}$ is unique. There can be multiple such $\vec{c}$.

Basic Feasible Solution

Definition

Let $P = \left\{\vec{x} \in \mathbb{R}^n | A\vec{x} \geq \vec{b}\right\}$ be a polyhedron. Let the constraint $\vec{a}_i \cdot \vec{x} \geq b_i$ be the constraint defined by the ith row of the matrix inequality. The constraint $\vec{a}_i \cdot \vec{x} \geq b_i$ is called **binding** or **active** at $\vec{x}^* \in \mathbb{R}^n$ if $\vec{a}_i \cdot \vec{x}^* = b_i$.

Remarks

1. If there are n linearly independent binding constraints at $\vec{x}^*$, then $\vec{x}^*$ is the unique solution to the linear system of equations defined by the n binding constraints. We will use this when proving things.

Definition

Let $P = \left\{\vec{x} \in \mathbb{R}^n | A_1\vec{x} \geq \vec{b}_1, A_2\vec{x} = \vec{b}_2\right\}$ be a polyhedron with both inequality constraints and equality constraints. Then

a) $\vec{x}^* \in \mathbb{R}^n$ is a **basic solution** of P if

1. All equality constraints are active
2. Out of all constraints active at $\vec{x}^*$, there exists a linearly independent subset of n of them.

b) If $\vec{x}^*$ is a basic solution that satisfies all constraints of P, then $\vec{x}^*$ is a **basic feasible solution (BFS)**.

Remarks

1. Any polyhedron can be written in terms of equalities and inequalities. We saw this when converting back and forth between equality and inequality constraints.
2. A basic solution need not lie in P.
3. We specifically need linearly independent constraints. If we have constraints like $0 \leq x \leq 3$ and $-1 \leq x \leq 4$, the second set of constraints is not as specific as the first set of constraints. We are still okay, since at most one of these four constraints can be active. This linearly independent requirement also allows us to still have reasonable definitions if the equality constraints are in the form of two inequality constraints.

Equivalence of the Definitions

Now for our big result!

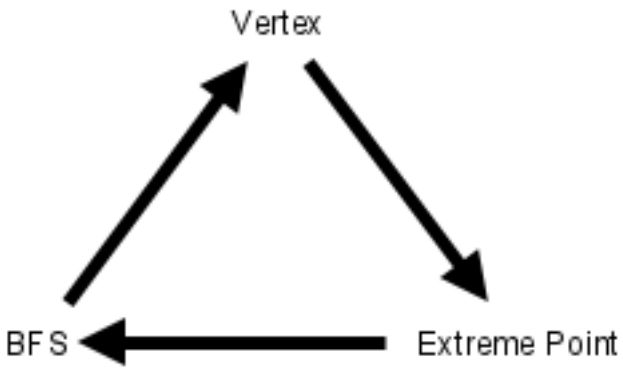
Theorem

In a polyhedron, P, a point is a vertex if and only if it is an extreme point if and only if it is a BFS. Symbolically:

$$\text{vertex} \iff \text{extreme point} \iff \text{BFS}$$

Proof

We will break the proof up into 3 pieces. Combining these gives the full equivalence.



Vertex $\implies$ Extreme Point

Let $\vec{x}^*$ be a vertex of P. Then this implies $\exists \vec{c} \in \mathbb{R}^n$ such that

$$\vec{c} \cdot \vec{x}^* < \vec{c} \cdot \vec{y}$$

for all $\vec{y} \in P$ such that $\vec{x}^* \neq \vec{y}$. Let $\vec{y}, \vec{z} \in P$ such that $\vec{y} \neq \vec{x}^*$ and such that $\vec{z} \neq \vec{x}^*$ and let $\lambda \in (0, 1)$. Then we know

$$\vec{c} \cdot \vec{x}^* < \vec{c} \cdot \vec{y}$$

$$\vec{c} \cdot \vec{x}^* < \vec{c} \cdot \vec{z}$$

and as $\lambda > 0$, we have

$$\lambda \vec{c} \cdot \vec{x}^* < \lambda \vec{c} \cdot \vec{y}$$

$$(1-\lambda)\, \vec{c} \cdot \vec{x}^* < (1-\lambda)\, \vec{c} \cdot \vec{z}$$

So if we add the constraints:

$$\vec{c} \cdot \vec{x}^* = \lambda \vec{c} \cdot \vec{x}^* + (1-\lambda)\, \vec{c} \cdot \vec{x}^* < \lambda \vec{c} \cdot \vec{y} + (1-\lambda)\, \vec{c} \cdot \vec{z} = \vec{c} \cdot (\lambda \vec{y} + (1-\lambda)\, \vec{z})$$

thus

$$\vec{x}^* \neq \lambda \vec{y} + (1-\lambda)\, \vec{z}$$

and thus $\vec{x}^*$ is an extreme point.

Extreme Point $\implies$ BFS

Instead of showing this directly, we will prove the contrapositive:

$$\text{Not BFS} \implies \text{Not Extreme Point}$$

Suppose $\vec{x}^* \in P$ is not a BFS. Let

$$S = \{\vec{a_i} \cdot \vec{x}^* = b_i\}$$

be the set of active constraints at $\vec{x}^*$ and let $I = \{i | \vec{a_i} \cdot \vec{x}^* = b_i\}$ be the set of indices of active constraints. As $\vec{x}^*$ is not a BFS, this implies there does not exist a set of n LI constraints as a subset of S. As such, we can find a vector $\vec{d} \in \mathbb{R}^n$ such that

$$\vec{a_i} \cdot \vec{d} = 0 \ \forall i \in I$$

Now, let $\vec{y} = \vec{x}^* + \epsilon \vec{d}$ and $\vec{z} = \vec{x}^* - \epsilon \vec{d}$ for some small $\epsilon > 0$ (we specify how small later). Then

$$\vec{a_i} \cdot \vec{y} = \vec{a_i} \cdot \left(\vec{x}^* + \epsilon \vec{d}\right) = \vec{a_i} \vec{x}^* + \epsilon \vec{a_i} \cdot \vec{d} = \vec{a_i} \vec{x}^* = b_i$$

Similarly,

$$\vec{a_i} \cdot \vec{z} = b_i$$

If $i \notin I$, (i.e. the corresponding constraint is not active), we see

$$\vec{a_i} \cdot \vec{x}^* > b_i \ \forall i \notin I$$

We would like

$$\vec{a_i} \cdot \vec{y} = \vec{a_i} \cdot \left(\vec{x}^* + \epsilon \vec{d}\right) = \vec{a_i} \cdot \vec{x}^* + \epsilon \vec{a_i} \cdot \vec{d} > b_i \ \forall i \notin I$$

so we can rearrange and see

$$\vec{a_i} \cdot \vec{x}^* - b_i > -\epsilon \vec{a_i} \cdot \vec{d}$$

As we can make epsilon arbitrarily small, and we know $\vec{a_i} \cdot \vec{x}^* - b_i > 0$, then we can pick ϵ such that

$$\vec{a_i} \cdot \vec{x}^* - b_i > \left|-\epsilon \vec{a_i} \cdot \vec{d}\right| = \epsilon \left|\vec{a_i} \cdot \vec{d}\right|$$

Suppose we pick ϵ such that

$$\epsilon \left|\vec{a_i} \cdot \vec{d}\right| < \vec{a_i} \cdot \vec{x}^* - b_i \ \forall i \notin I$$

Then

$$\vec{a_i} \cdot \vec{x}^* + \epsilon \vec{a_i} \cdot \vec{d} > b_i \ \forall i \notin I$$

Similarly, we have

$$\vec{a}_i \cdot \vec{z} = \vec{a}_i \cdot \vec{x}^* - \epsilon \vec{a}_i \cdot \vec{d} > b_i \ \forall i \in I$$

if we use the same ϵ. As $\vec{y}, \vec{z}$ both satisfy all non-active constraints, then $\vec{y}, \vec{z} \in P$. This means we can write

$$\frac{1}{2}\vec{y} + \frac{1}{2}\vec{z} = \frac{1}{2}\left(\vec{x}^* + \epsilon\vec{d}\right) + \frac{1}{2}\left(\vec{x}^* - \epsilon\vec{d}\right) = \vec{x}^*$$

so $\vec{x}^*$ is not an extreme point.

BFS $\implies$ Vertex

Let $\vec{x}^*$ be a BFS and let $S = \{\vec{a_i} \cdot \vec{x}^* = b_i\}$ be the set of all binding constraints, and let $I = \{i | \vec{a_i} \cdot \vec{x}^* = b_i\}$. Let

$$\vec{c} = \sum_{i \in I} \vec{a}_i$$

then for $\vec{x}^*$, we have

$$\vec{c} \cdot \vec{x}^* = \left(\sum_{i \in I} \vec{a}_i\right) \cdot \vec{x}^* = \sum_{i \in I} \vec{a}_i \cdot \vec{x}^* = \sum_{i \in I} b_i$$

but as there are n linearly independent active constraints, we know that $\vec{x}^*$ is the unique point in $\mathbb{R}^n$ with this property. As such, any other point $\vec{y}$ in P must have at least one $i \in I$ such that $\vec{a}_i \cdot \vec{y} > b_i$. Therefore, for all other $\vec{y} \in P$, we see

$$\vec{c} \cdot \vec{y} = \left(\sum_{i \in I} \vec{a}_i\right) \cdot \vec{y} = \sum_{i \in I} \vec{a}_i \cdot \vec{y} > \sum_{i \in I} b_i = \vec{c} \cdot \vec{x}^*$$

Thus $\vec{x}^*$ is a vertex.